p4-move-zip-fixer
Technical Brief for Perforce Engineering Review

Audience	Perforce Support and Engineering
Status	Prototype implementation, MIT licensed. Seeking technical review prior to customer deliver
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Related Case	P4 Migration Issue - Case# 01565143

Overview

This document describes a prototype utility developed to assist with a migration issue observed during large-scale use of p4 zip. The tool was created specifically to help address cases where p4 zip aborts because a changelist contains move/add or move/delete actions whose counterpart files are located outside the command view.

Problem Statement

The currently recommended workaround, namely widening the remote spec to include both sides of every move operation, is technically correct. However, at the scale of the Amdocs depot, which contains approximately 705,000 unique paths and decades of accumulated history, manual enumeration becomes operationally impractical.

Approach

The prototype implements a three-stage workflow based entirely on the existing p4 command set. No server modifications, protocol changes, or privileged access are required.

Workflow:
Customer Depot -> scan phase using p4 filelog -ztag -> build-spec phase -> p4 zip execution.

The scan phase operates in parallel and supports resumable execution. Results are cached in a local SQLite database to allow interrupted runs to continue without restarting from scratch.

Conservative Design Principles

Key characteristics include read-only interaction with the live depot, no depot mutation operations, resumable execution using SQLite state tracking, and fail-fast behavior with changelist extraction from p4 zip failures.

Questions for Perforce Engineering

1. Does filelog -ztag reliably expose both sides of a move when only one side falls within the supplied path filter?

2. Are there known server configuration flags or historical behaviors that alter how movedFile values are reported?

3. How are chained move sequences represented across changelists, for example A -> B -> C?

4. For branched-then-moved files, should additional integration records be traversed?

5. Is there a more efficient bulk-query strategy for very large depots than p4 filelog -c <range> //depot/... ?

Current Scope and Limitations

This prototype is not intended as a replacement for the official enhancement request currently under discussion. It is not officially supported software and has not yet been validated against all server versions.

Repository Structure

The repository includes a parallel filelog scanner, SQLite-backed resumable cache, remote spec generator, p4 zip wrapper and parser, CLI interface, and unit tests using mocked P4 interfaces.

Requested Feedback

We would appreciate a short technical review call, guidance on the open questions above, and feedback regarding whether Perforce is comfortable with the tool being shared with Amdocs.

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